COSC2731 - Tutorial 1

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RMIT UNIVERSITY

Outline

1. Warm greeting
2. Question 1
3. Question 2
4. Question 3
5. Question 4



Warm greeting

- Name: **Tuan Phong Ngo**
- Email: phong.ngo@rmit.edu.vn
- Room: Hanoi Campus, Tower A, Level 5
- Profile:
  - Ph.D. in Computer science, Sweden
  - Experienced as an IT lecturer and senior compiler engineer
- Tutorials:
  - 8:30-10:00, Wednesday, HN 1.3.B006
- Consultations:
  - 10:00-11:00, Friday, HN 1.5.B027





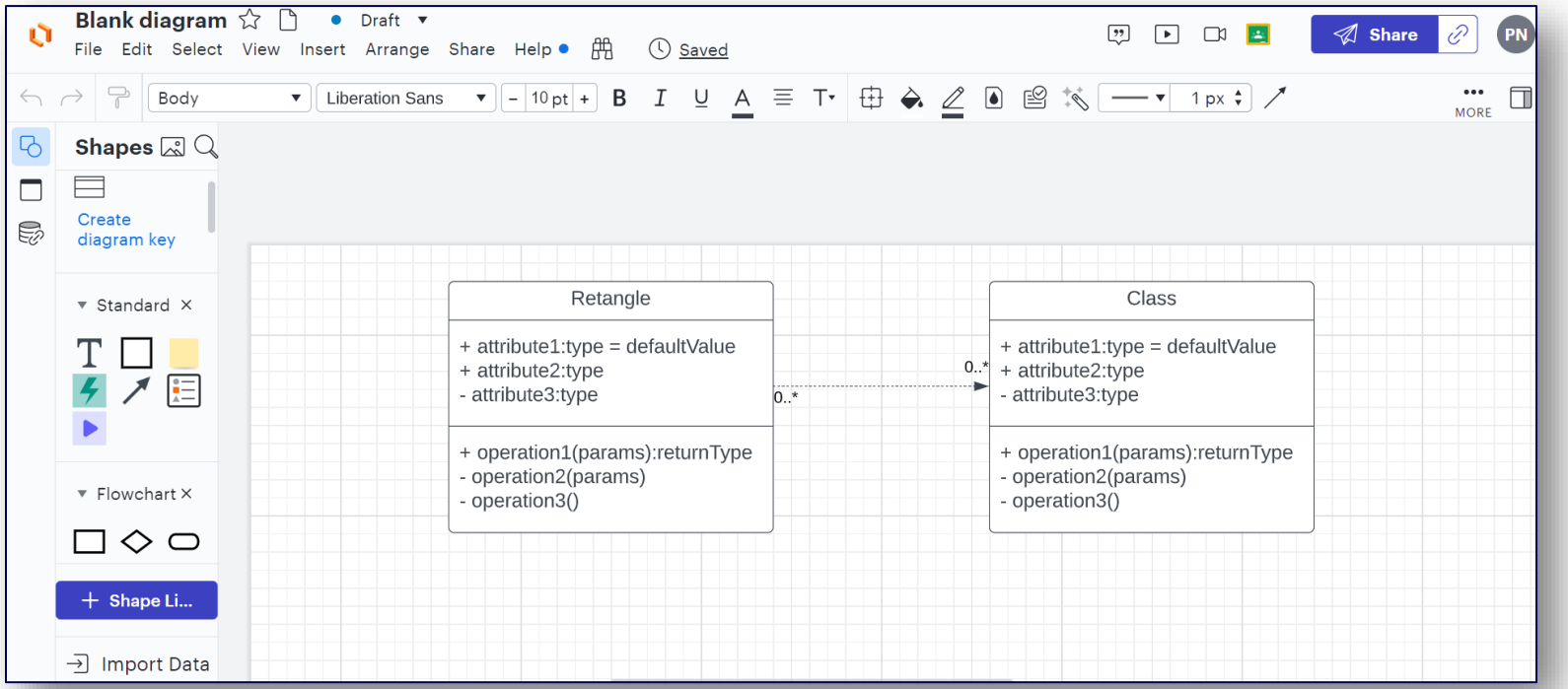




How about you (name, hobbies, programming experience etc.)?

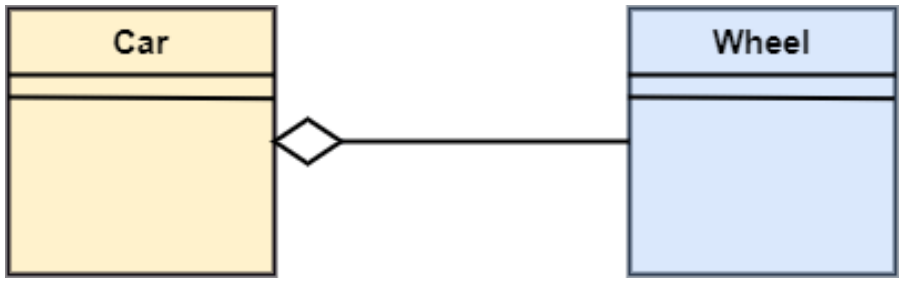
Question 1 – UML tools

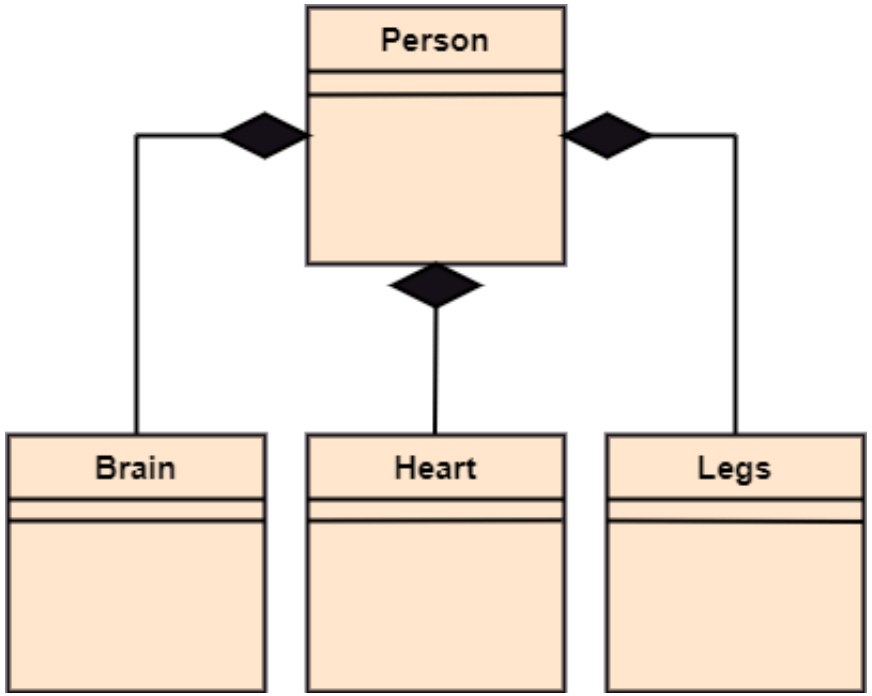
- Let's get familiar with some basic UML class diagram notations: https://jairanjankumar.gitbook.io/cheatsheet/uml-class-diagram
- If you draw UML diagrams on a computer, you can use https://www.lucidchart.com/pages



Lucidchart interface

Aggregation vs. Composition



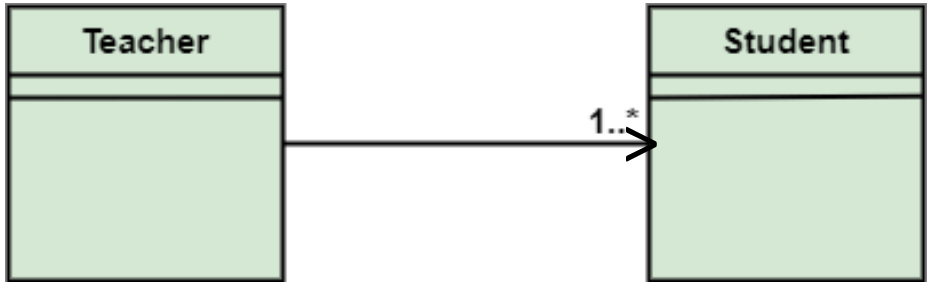


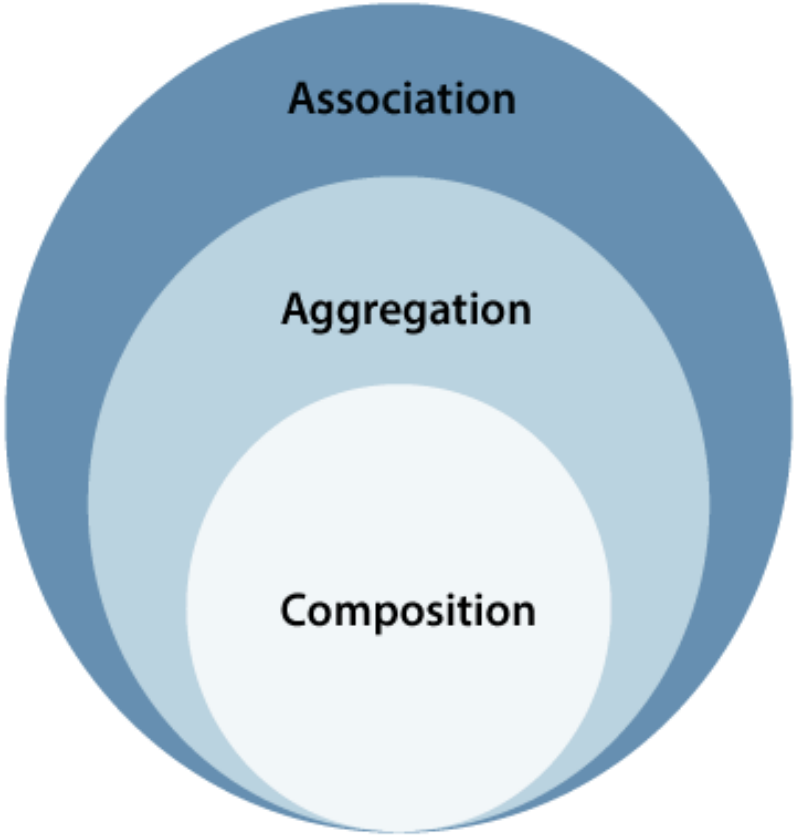
Aggregation: in a *part-of relationship*, the child *can* exist independently of the parent

Composition: in a *part-of relationship,* the child *cannot* exist independent of the parent

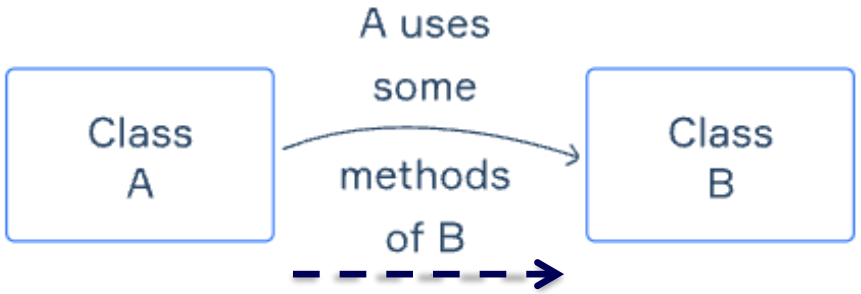
https://www.javatpoint.com/uml-association-vs-aggregation-vs-composition

Association and Dependency





Association: one class has another as an attribute



Dependency: a class uses another by parameter or return type

Question 2 – Draw UML diagram

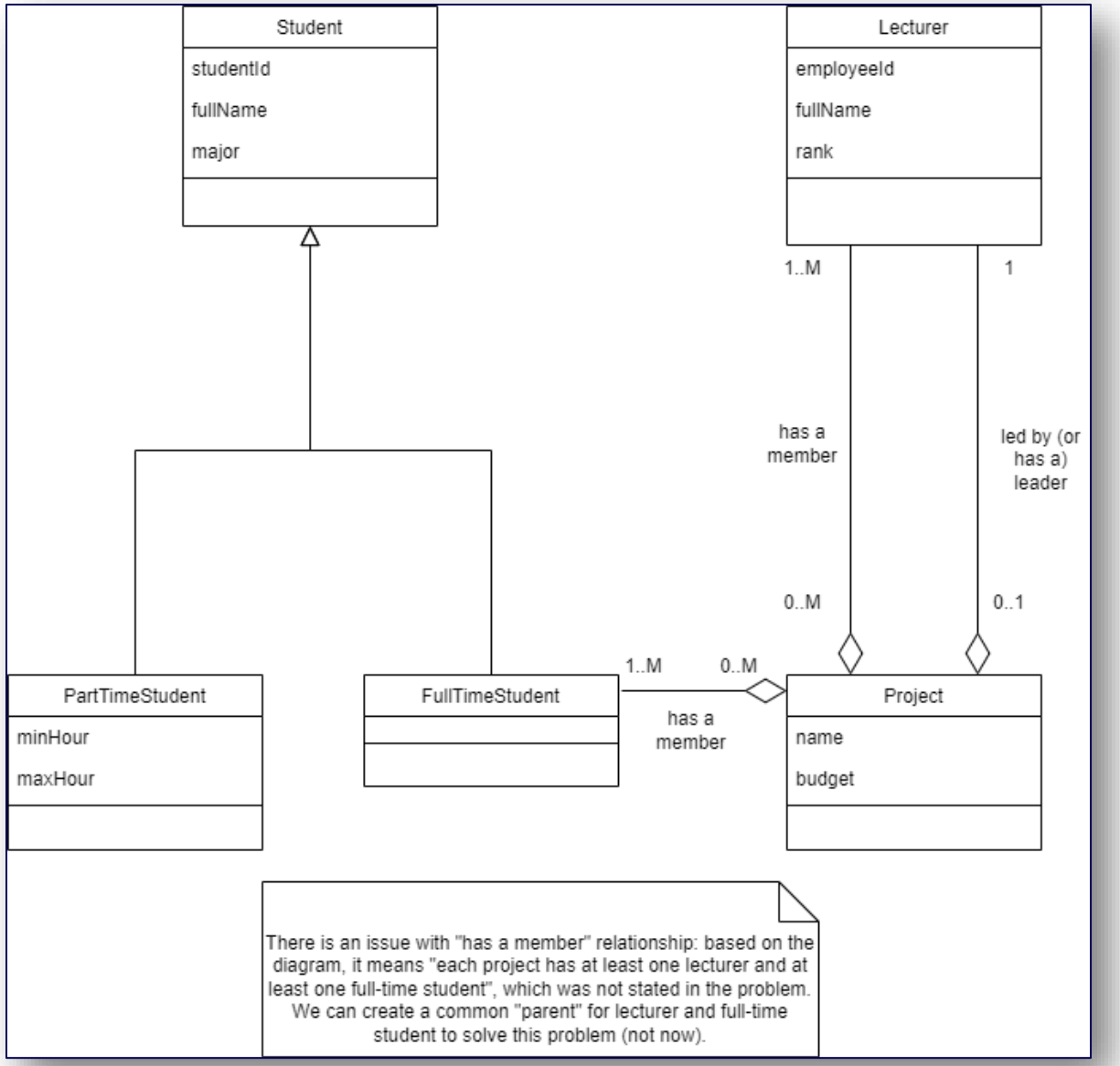
- RMIT University has two types of **students**: **full-time** and **part-time**. All students have student id, full name, and major. Moreover,
  - Only full-time students can optionally join research projects.
  - Part-time students must register the minimum and maximum hours they spend on instructor-led learning.

- There are **lecturers**, who are responsible for teaching and research. Each lecturer has an employee id, full name, and rank. Each lecturer can lead at most one research project at a time, but she/he can join many projects at the same time.

- Each research **project** has a name, a budget, one leader (who must be a lecturer), and one or more members (each can be a full-time student or a lecturer).

Let's take 10 mins for drawing your UML diagram

Question 2 - Discussion



- Does the diagram say about "**A project has one or more members (each can be a full-time student or a lecturer)**"? How?



Question 3 – Implement Student class

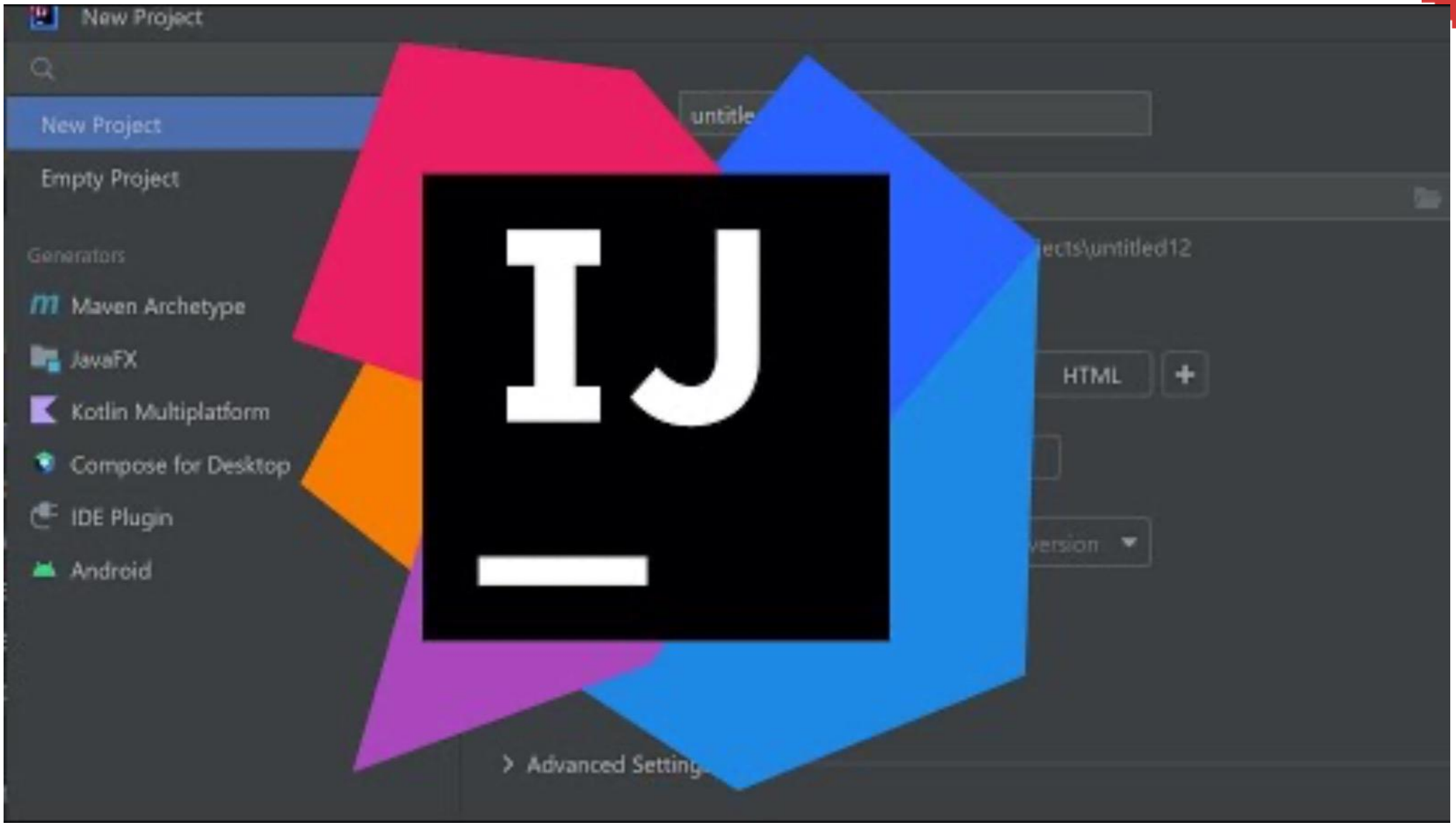


- Implement the class **Student**. Decide the appropriate access modifiers for the attributes.

- Implement a class **StudentTest** in the **same package** as the Student class. In StudentTest, use the main() method to instantiate Student objects and access/modify their attributes.

- Create a class **StudentTest2** in a **different package**. Create Student objects and access/modify their attributes from this class.

How to create and run a Java project on IntelliJ



https://www.youtube.com/watch?v=Y95E0W5wfFE

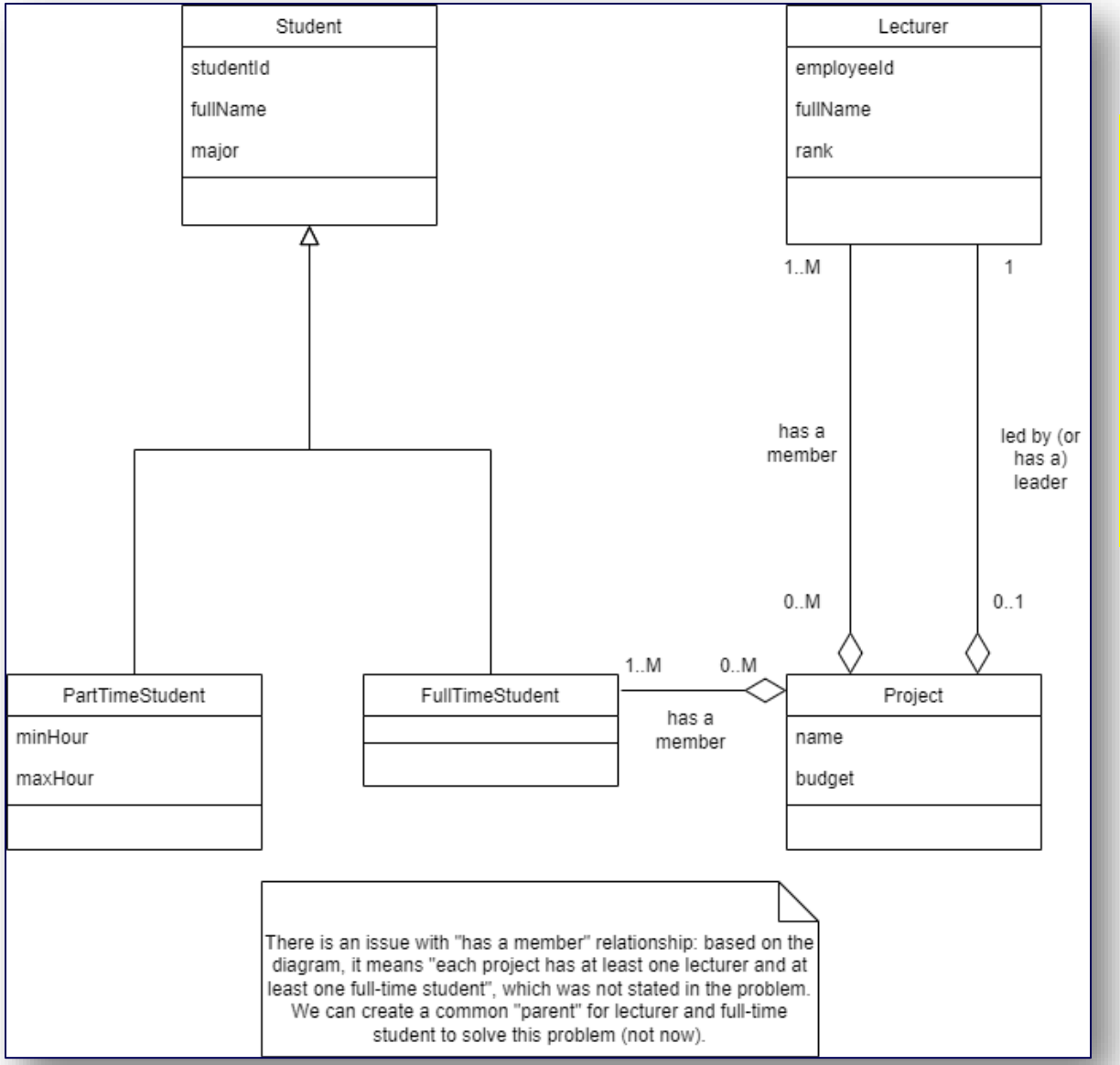
Question 4 – Implement Project and Lecturer class

- In **Lecturer** class, implement two methods:
  - **boolean leadProject(Project p)**: returns true if the current Lecturer object can lead the project specified in the parameter and return false otherwise.
  - **boolean joinProject(Project p)**

- In **Project** class, implement a method:
  - **void displayMembers()**: prints out all members of the current project (including its leader and regular members).

- Use **Javadoc** for the comments

Question 4 - Discussion



- How should we implement "**has a leader**" relationship between Lecturer and Project?
- How should we implement "**has a member**" relationship between Lecturer and Project?

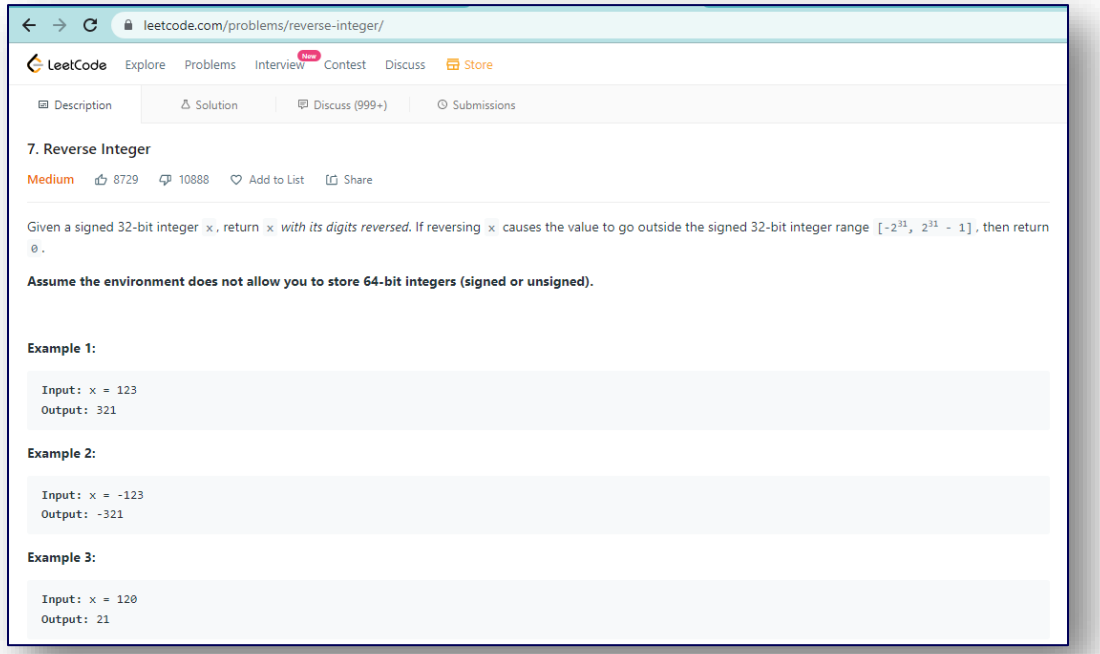


Learning at home

Review lectures

Finish this week tutorial and extra exercises

Optional: Solve some easy/medium problems at https://leetcode.com/problemset/all/



Have fun coding &
See you next week

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